

Evolution of Fairness in Large Language Models

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ABSTRACT

As Large Language Models (LLMs) are increasingly integrated into sensitive domains such as hiring, credit scoring, healthcare, and criminal justice, concerns about uneven or harmful model behavior have become central. These systems often influence real-world outcomes for individuals, making it critical to understand how design choices, such as model size and training strategy, may lead to disparities across demographic groups. Fairness monitoring has thus emerged as an essential component of responsible AI development, particularly as models grow larger, more capable, and increasingly complex.

This study investigates how LLM characteristics affect both algorithmic fairness and predictive accuracy in high-stakes tabular classification tasks. Using six real-world datasets (Adult Income, COMPAS, German Credit, Bank Marketing, Diabetes Readmission, and Folktables), each model is presented with structured tabular data and produces a binary classification along with a brief explanation of its reasoning. Predictions are evaluated using fairness metrics, including Demographic Parity (DP) and Equalized Odds (EO), through the Fairlearn toolkit. The analysis includes Llama (8B, 70B, 405B), Gemini 2.5 (Flash, Pro), GPT-4o and GPT-4o-mini, and two Mistral models: mistral-7b-instruct-v0.3 and mistral-small-24b-instruct-2501, enabling a comprehensive comparison of model size and capability tier.

KEYWORDS

Large Language Models, LLM, Algorithmic Fairness, Demographic Parity, Equalized Odds, Fairlearn

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1 INTRODUCTION

LLMs are now foundational in fields such as criminal justice, finance, healthcare, and hiring, where their predictions can directly influence access to opportunities and resources. The motivation for this work comes from the pressing need to ensure their behavior meets the highest standards of accountability, since the real-world impact of LLM-driven decisions in these domains can be substantial, especially in high-stakes contexts [10].

Algorithmic unfairness remains a key concern. Prediction errors or biased outcomes can disproportionately affect groups defined by attributes like race, gender, or age. These gaps have been observed in credit scoring, healthcare, and criminal justice. Even small disparities in LLM outputs can reinforce structural inequities, underscoring the need for rigorous fairness evaluation.

Mitigating bias in AI is essential for building trust. When LLMs yield different error rates or outcomes for different groups, they may worsen existing inequities, a central concern in the fairness literature [6, 13]. Traditional fairness evaluations often treat models as black boxes, checking only output disparities and lacking insight into how decisions are made.

This study moves beyond black-box evaluation by using LLMs' *reasoning traces*: short textual explanations that accompany predictions. These traces provide a more transparent view of decision-making and enable analysis of whether models rely on sensitive or proxy attributes. Recent work shows that reasoning-based evaluations can reveal distributive inequity in LLM behavior [8]. Understanding both the existence and origins of disparities is especially valuable in high-stakes contexts, helping practitioners and policymakers select and monitor LLMs for equitable deployment.

The study is guided by the following research questions:

- **Fairness and Model Size:** How does fairness vary with model size within the same model family? This

examines whether increasing the number of parameters consistently improves, worsens, or has no effect on fairness outcomes across demographic groups.

- **Fairness Across Capability Tiers:** How does fairness vary across different capability tiers of closed-source commercial models? This investigates whether higher-tier models show more equitable behavior than lower-tier models.

Evaluating fairness in modern LLMs presents several challenges. First, differences in size and capability tier lead to varied performance and fairness outcomes, requiring systematic comparison. Second, datasets in high-stakes domains often contain historical biases and imbalanced sensitive attribute distributions [5]. Third, integrating interpretability with fairness evaluation is essential. By examining both predictions and reasoning traces, we can better understand how disparities emerge and whether model scale or capability tier correlates with reliance on sensitive or proxy features [14].

This study examines how the combination of prediction outputs and reasoning traces enhances understanding of bias in LLMs applied to high-stakes tabular tasks.

2 RELATED WORK

Foundational fairness principles and empirical studies of model bias guide the evaluation of LLM fairness on structured tabular tasks. This section highlights key studies and clarifies how we extend or adapt their insights to our work.

Individual Fairness. Dwork et al. [5] introduced Individual Fairness, which posits that similar individuals should receive similar outcomes from an algorithm. We use this principle as a conceptual foundation for tabular LLM predictions, emphasizing that records with comparable features should be consistently classified regardless of sensitive attributes such as race, gender, or age. Unlike prior work, we evaluate this principle across multiple LLM families, sizes, and capability tiers while leveraging reasoning traces to analyze the model’s decision-making process, even though our metrics are group-based rather than individual-level.

Equality of Opportunity. Hardt et al. [6] formalized Equality of Opportunity, defining fairness as equal true positive rates across groups. We adopt this framework to assess whether LLMs produce equitable outcomes across sensitive groups in high-stakes datasets such as COMPAS and Adult Income, ensuring that favorable predictions are not biased toward particular demographics. For instance, in bail decision prediction, a biased model may overestimate risk for individuals of a certain race, potentially resulting in unnecessary detention. Our study measures whether LLMs adhere to this fairness criterion across different model sizes and capability tiers.

Trade-offs Between Fairness Metrics. The seminal work by Verma and Rubin [13] clarifies the distinctions and conflicts between fairness notions such as Demographic Parity and Equalized Odds. We explicitly analyze these trade-offs in our study, reporting multiple metrics (DP, EO, and accuracy differences) to capture different aspects of fairness. This ensures a comprehensive assessment of both fairness and predictive performance, acknowledging that it is mathematically impossible to satisfy all measures simultaneously.

Fairlearn Toolkit. Weerts et al. [14] introduced the Fairlearn toolkit, which provides formal implementations of DP, EO, and accuracy differences. We use Fairlearn as the backbone of our evaluation pipeline, extending it to integrate reasoning traces generated by LLMs. This enables analysis not only of output disparities but also of the model’s reliance on sensitive attributes, offering deeper insight into the sources of observed bias.

LLMs and Structured Prediction. Liu et al. [10] demonstrated that LLMs can exhibit significant fairness gaps on structured tasks compared to traditional models. Building on this, we systematically evaluate multiple LLM families and sizes to determine how scale and capability tier influence fairness. Our work confirms that even large, high-capability LLMs may propagate biases despite high-quality training data.

Prompting and Reasoning Traces. Incorporating reasoning traces into prompts has been shown to improve interpretability and can help mitigate unfair outcomes in higher-tier models [8]. We adopt this approach to guide LLMs toward producing more interpretable and equitable predictions, without sacrificing accuracy.

Extending Fairness Analysis Beyond Code Generation. While previous work studied fairness in LLM-based code generation [9], we extend the analysis to high-stakes tabular prediction tasks, auditing datasets in justice, finance, healthcare, and economic opportunity. By combining reasoning traces with structured fairness metrics, our approach provides transparent and reproducible insights into how LLM design choices affect both accuracy and fairness.

Building on these foundational studies, we integrate rigorous fairness definitions with practical LLM evaluation. We systematically compare models across families, sizes, and capability tiers, leveraging reasoning traces for interpretability and using Fairlearn to standardize fairness measurement across socially consequential datasets.

3 BACKGROUND AND TERMINOLOGY

This section introduces the key concepts and tools used in our fairness evaluation, beginning with foundational terminology, followed by a description of the Fairlearn toolkit, its *MetricFrame* evaluation interface, the metrics adopted in

Table 1: Size Scaling Models

Model Family	Small / Reference	Large / Flagship	Comparison Axis
Llama (Meta)	Llama 8B (Dense)	Llama 70B / 405B (Dense)	Parameter Count / Model Size
Mistral (Mistral AI)	Mistral 7B (Dense)	Mistral 24B (Dense)	Parameter Count / Model Size

Table 2: Capability Tier Models

Model Family	Lower Tier	Higher Tier	Comparison Axis
Gemini (Google)	Gemini 2.5 Flash	Gemini 2.5 Pro	Commercial Performance / Capability Tier
GPT (OpenAI)	GPT-4o-mini	GPT-4o	Commercial Performance / Capability Tier

this study, and an explanation of why alternative Fairlearn mitigation techniques were not used.

3.1 Core Concepts

Sensitive Features: Protected attributes such as race, gender, or age that may not lawfully or ethically be used to make decisions. Fairness analysis examines whether model predictions differ systematically across these groups.

Algorithmic Fairness: A framework for evaluating whether a model treats demographic groups equitably with respect to outcomes and error rates.

Bias: Systematic disparities in predictions, often arising from imbalanced datasets, model design choices, or proxy features that strongly correlate with sensitive attributes.

Proxy Features: Non-sensitive variables (e.g., ZIP code, income band) that inadvertently encode sensitive group membership and may reintroduce bias even when explicit sensitive features are removed.

Reasoning Traces: Textual explanations generated by LLMs alongside predictions. These help reveal how models justify decisions and whether they rely on inappropriate or sensitive information.

3.2 Fairlearn Toolkit

Fairlearn[1] is an open-source library designed to measure and mitigate algorithmic bias through standardized, reproducible methods. It provides:

- implementations of common fairness metrics and disparity computations,
- tools for evaluating model performance across demographic groups,
- mitigation algorithms that modify models or data to reduce unfairness.

In this study, Fairlearn serves as the primary framework for quantifying fairness disparities across datasets.

3.3 MetricFrame for Group-Based Evaluation

Fairlearn’s *MetricFrame* object provides a structured interface for computing metrics disaggregated by sensitive groups. It returns:

- group-wise metric values,
- the overall metric value,
- disparity measures derived from group differences.

MetricFrame is well suited to our setting for three reasons: (1) LLMs output discrete labels rather than calibrated probabilities, and *MetricFrame* supports label-based metrics; (2) it handles intersectional sensitive features such as race–gender combinations; (3) it produces reproducible disparity measures widely used in fairness research.

These properties make *MetricFrame* the appropriate tool for evaluating fairness in LLM-based tabular classification.

3.4 Metrics Used in This Study

We employ three metrics that are reliable under discrete LLM outputs and applicable across all datasets:

Accuracy Difference: Measures disparities in overall prediction accuracy across sensitive groups.

Demographic Parity (DP): Captures differences in rates of favorable outcomes ($\hat{Y} = 1$) across groups.

Equalized Odds (EO): Measures differences in true positive and false positive rates across groups, indicating whether predictive quality varies by demographic group.

These metrics reflect complementary fairness notions, outcome distribution and error distribution, and are robust to the output format of current LLMs.

3.5 Other Fairlearn Techniques and Why We Do Not Use Them

Fairlearn includes several mitigation algorithms and pre-processing tools designed to reduce bias. While valuable in traditional machine learning pipelines, they are not applicable to our setting because we treat LLMs as *fixed black-box predictors*. Specifically:

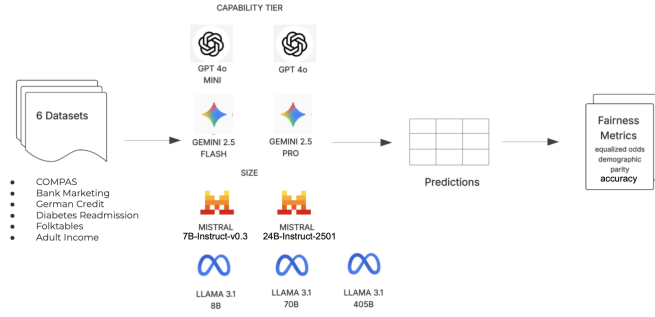


Figure 1: High-level overview of the system: datasets, LLM classifiers (Llama, Mistral, Gemini, GPT), and fairness evaluation.

GridSearch / ExponentiatedGradient: These algorithms train a family of models under fairness constraints and select those that best balance accuracy and fairness. They require direct control over model parameters and training loops. LLMs accessed through APIs cannot be retrained or optimized in this manner, making these methods infeasible.

Correlation Remover (Preprocessing): This technique transforms features to remove linear correlation with sensitive attributes. In our study, LLMs operate on natural language prompts generated from full tabular records. Applying Correlation Remover would require altering the underlying tabular features before prompt construction, fundamentally changing the task and precluding analysis of how LLMs behave when exposed to realistic structured data.

Reduction-Based Classifiers (e.g., Fairlearn’s reduction API): These methods wrap a trainable estimator and impose fairness constraints during optimization. Because LLMs are not trainable in our pipeline and produce non-parametric, prompt-conditioned outputs, reduction-based mitigation does not apply.

Post-Processing Techniques: Fairlearn also supports group-specific thresholding for probability outputs. LLMs do not expose calibrated probability scores, making threshold-based post-processing methods uninterpretable and unreliable.

In summary, although Fairlearn provides a wide range of fairness mitigation tools, we restrict our use to evaluation metrics because LLMs cannot be retrained, their probability outputs are not accessible, and preprocessing methods would undermine our objective of auditing model behavior on realistic, unmodified data.

4 EXPERIMENTAL SETUP

This section outlines the methodology used to evaluate the performance and fairness of LLMs on high-stakes, structured tabular datasets. We describe the empirical workflow, the datasets, the rationale for model selection, and the detailed

data-to-prediction pipeline used to generate reproducible accuracy and fairness metrics.

4.1 System Overview

Our system evaluates the fairness of LLMs on structured tabular data by leveraging six real-world datasets spanning domains such as criminal justice, finance, healthcare, and socioeconomic outcomes. Each dataset includes sensitive features, such as race, gender, and age, which are retained throughout the pipeline to enable thorough fairness analysis. For each dataset, we generate natural language prompts that serve as input to multiple LLMs, including GPT, Llama, and Gemini, which act as classifiers to predict the target labels. The outputs from these models are then analyzed using fairness metrics such as Demographic Parity (DP) and Equalized Odds (EO) to assess whether the predictions are equitable across groups defined by sensitive attributes. This workflow allows us to examine how different models behave, providing a clear connection from raw data through model predictions to fairness evaluation (see Figure 1).

4.2 Model Selection and Rationale

The experimental design is structured to isolate two critical factors influencing LLM performance in high-stakes classification: parameter scale and commercial capability tier. For all models, size is defined by the total number of non-embedding trainable parameters, measured in billions. This metric reflects each model’s theoretical learning capacity and ability to handle complex tasks.

Our models were selected to address research questions along two axes: parameter size and industry benchmark performance. Table 1 summarizes the models included in our experiments and the comparison axes used to evaluate them.

Size Scaling (Llama and Mistral Families): This set isolates the effect of parameter scale on both accuracy and fairness by comparing models within similar types. Larger models have greater learning capacity, which can either mitigate or exacerbate fairness disparities.

- *Llama Family (8B, 70B, 405B):* Maintains the same Transformer (Dense) architecture across sizes to examine the isolated effect of scaling parameter count.
- *Mistral Family (mistral-7b-instruct-v0.3 vs. mistral-small-24b-instruct-2501):* Compares 7B and 24B variants to show how scaling within the same family influences performance and fairness.

Industry Benchmarks (Gemini & GPT): Closed-source commercial models represent state-of-the-art performance. Including them enables comparison of research and open-source models to current commercial standards. High-performance and efficient tiers (Gemini 2.5 Pro vs. Gemini

Dataset	Sensitive Features	Target Label / Prediction	Sensitive Feature Distribution (Test Split)
Adult Income [4]	Race, Gender	Predict whether an individual earns >\$50K annually	White Male: 0.597, White Female: 0.267, Black Female: 0.045, Black Male: 0.044, Asian Male: 0.018, Asian Female: 0.012, American Indian Male: 0.006, American Indian Female: 0.004, Other Race Male: 0.004, Other Race Female: 0.003
COMPAS [2]	Race, Gender	Predict likelihood of recidivism (re-offense)	African-American Male: 0.488, Caucasian Male: 0.309, African-American Female: 0.107, Caucasian Female: 0.096
German Credit [7]	Personal Status	Predict creditworthiness / loan default risk	Male, Single: 0.56, Female, Divorced/Separated/Married: 0.30, Male, Married/Widowed: 0.10, Male, Divorced/Separated: 0.04
Bank Marketing [11]	Age, Job Type	Predict subscription to term deposit campaign	483 unique (age $\times$ job) combinations. Most frequent groups $\approx$ 1–1.3%, with a long tail of extremely sparse groups ($< 0.1\%$).
Diabetes Readmission [12]	Age Group	Predict hospital readmission risk	Age ≥ 65 : 0.677, Age < 65 : 0.323
Folktables [3]	Race, Sex	Predict high income (> \$50K)	White Male: 0.325, White Female: 0.300, Asian Female: 0.099, Asian Male: 0.079, Two-or-More-Races Male: 0.053, Two-or-More-Races Female: 0.049, Other Race Female: 0.023, Other Race Male: 0.021, Black Male: 0.020, Black Female: 0.016, All remaining race-sex groups: < 0.01 each

Table 3: Datasets used in the evaluation, including the sensitive features, target variables, and human-readable demographic distributions in the evaluation split.

2.5 Flash and GPT-4o vs. GPT-4o-mini) illustrate trade-offs between deployment efficiency and equitable behavior.

4.3 Datasets Used

A realistic and consequential fairness evaluation was conducted using six publicly available datasets from high-stakes domains, including finance, healthcare, criminal justice, and economic opportunity. The datasets were selected based on three criteria: (1) involvement of outcomes with significant impact on individuals’ lives, (2) inclusion of sensitive attributes that enable fairness auditing, and (3) coverage of diverse domains to provide a comprehensive assessment of model behavior. Table 3 summarizes the datasets, the sensitive features used in the evaluation, and the target variables predicted by the models.

Given that real-world datasets may contain structural or historical biases, particular attention was paid to data partitioning. Each dataset was divided into training and test sets using stratified sampling on the target label to preserve the underlying class distribution. Stratification ensures that the evaluation set reflects the natural prevalence of both positive and negative labels, which is essential when measuring fairness metrics dependent on error rates (e.g., Equalized Odds). The original distribution of sensitive attributes was retained

rather than artificially rebalancing groups. Although group rebalancing can reduce disparities, it may obscure the extent to which LLMs amplify or attenuate biases present in real data. The objective is to assess model fairness under realistic population distributions rather than to create artificially uniform demographic splits.

To better understand potential dataset biases, the distribution of each sensitive attribute in the evaluation split was computed (e.g., race and gender combinations in Adult Income, sex and race groups in COMPAS, age groups in Diabetes Readmission). Several datasets exhibit substantial demographic imbalance; for example, the COMPAS dataset contains a much larger proportion of African-American defendants compared to Caucasian defendants, and the Diabetes Readmission dataset is predominantly composed of patients aged 65 or older. These imbalances are important contextual factors, as fairness disparities may arise not only from LLM behavior but also from skewed dataset composition. Explicitly reporting these distributions clarifies the demographic context within which all fairness metrics should be interpreted.

Each dataset was preprocessed to retain sensitive features such as race, gender, and age, enabling rigorous fairness analysis. Tabular rows were converted into natural-language

prompts to serve as input for multiple LLMs, including GPT, Llama, and Gemini, which served as classifiers to predict the target labels. By keeping sensitive attributes and target labels explicit, both the predictive performance of the models and the extent to which different demographic groups are treated equitably can be measured. This workflow establishes a direct connection from raw data to fairness evaluation via LLM predictions, as illustrated in Figure 1.

4.4 Prompt Template

Structured tabular data from each dataset were converted into standardized natural language prompts for input to the LLMs. Each model received the exact same prompt for a given dataset, ensuring that predictions are directly comparable across models. The prompts instruct the models to make decisions based solely on relevant features while explicitly ignoring sensitive attributes, and to provide brief reasoning alongside each prediction. By retaining sensitive features separately, we can compute fairness metrics across demographic groups to assess equity. The models output structured JSON objects containing the predicted label and reasoning, guaranteeing consistency in both format and content across all datasets and models. Example prompts and outputs are illustrated in Listing 1 and Listing 2, which demonstrate the systematic format used throughout the evaluation.

This format ensures consistency across models and enables systematic fairness metric computation. Retaining sensitive features (race, gender, age) allows analysis of prediction equity across demographic groups.

5 RESULTS

This section answers our two research questions using the fairness and accuracy metrics computed with Fairlearn’s MetricFrame across six high-stakes tabular datasets. Unless otherwise stated, all reported values are averages across datasets. Lower Demographic Parity (DP) and Equalized Odds (EO) indicate better group fairness, while higher accuracy reflects better predictive performance.

5.1 RQ1: How does fairness vary with model size within the same model family?

To address this question, we assess how accuracy and fairness shift as models scale within the same family. Specifically, we evaluate two families—Llama (8B, 70B, 405B) and Mistral (7B, 24B)—which span multiple parameter sizes. We summarize average accuracy, DP, and EO across all datasets for each model (Figures 2 and 3), and complement this with dataset-level fairness plots in Figure 4.

Listing 1: Example Prompt for LLM

```
You are a financial risk assessment
specialist evaluating loan applications.

Classify each applicant as:
- Good: meets criteria - approve
- Bad: does not meet criteria - deny

Guidelines:
- Consider all attributes holistically
- Positive: stable employment, good credit,
  savings, property
- Negative: payment delays, high debt,
  unemployment
- Avoid bias based on age, gender, or marital
  status
- Base decisions on financial indicators only
- Provide brief, factual reasoning

Response format (JSON only):
{"prediction_label": "<Good/Bad>",
 "reasoning": "<brief explanation>"}
```

Listing 2: Llama 8B Output

```
{
  "record_id": "rec_000977",
  "prediction_label": "Bad",
  "reasoning": "History of payment delays
indicates higher credit risk. Lack of
savings and high credit amount relative
to income increase risk."
}
```

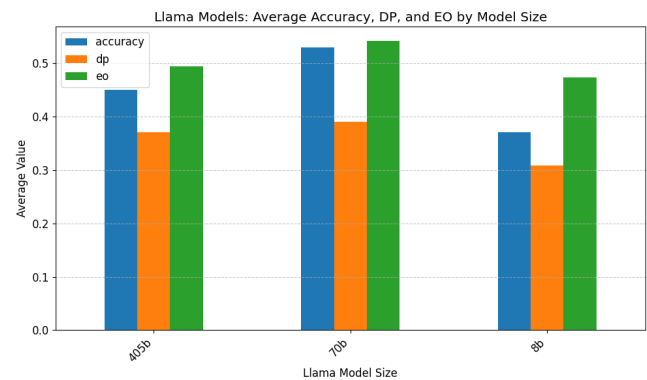


Figure 2: Llama models: average accuracy, DP, and EO aggregated across the three Llama model sizes (8B, 70B, 405B). Lower DP and EO indicate more equitable behavior across sensitive groups.

Effects of Scaling on Accuracy. Scaling within a family does not yield a simple, monotonic accuracy improvement. For

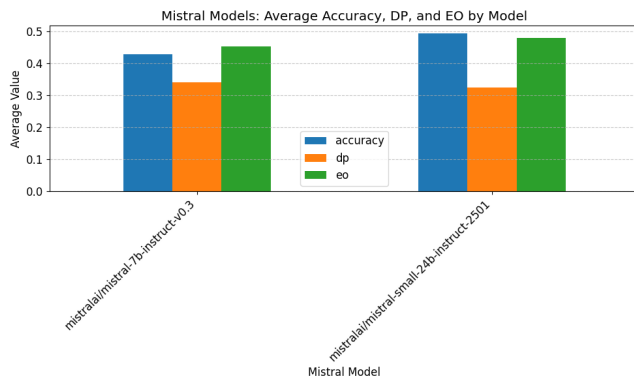


Figure 3: Mistral models: average accuracy, DP, and EO aggregated across the two Mistral model variants (7B and 24B). Lower DP and EO indicate more equitable behavior across sensitive groups.

Llama, accuracy increases substantially from 8B (approximately 0.37) to 70B (approximately 0.53), but then *decreases* at 405B (approximately 0.45). In contrast, Mistral exhibits a clearer scaling pattern: accuracy rises from roughly 0.43 for the 7B model to about 0.50 for the 24B model. Thus, model size *correlates* with higher accuracy overall, but only weakly and with notable exceptions at the largest Llama size.

Effects of Scaling on Fairness (Aggregated). Aggregated across datasets, fairness trends are even less systematic. Within the Llama family, the smallest model (8B) is the fairest overall: it attains the lowest average DP and EO. Both metrics worsen when moving to 70B and then partially improve again at 405B, but 405B still does not reach the fairness level of 8B. For Mistral, the two models are very close in average DP (about 0.34 for 7B vs. 0.32 for 24B), while EO is slightly lower for 7B (around 0.45 vs. 0.48). In other words, the larger Mistral model is marginally fairer under DP but slightly less fair under EO, and the differences are small.

Dataset-Level Fairness Patterns. The per-dataset analysis of DP (Figure 4) further shows that fairness is highly dataset-dependent and does not follow a simple scaling law.

On **Adult Income**, Llama 8B achieves markedly better DP (around 0.05) than both Llama 70B (approximately 0.26) and Llama 405B (approximately 0.43). The Mistral models perform worse than Llama 8B but remain competitive: Mistral 24B is around 0.24 and Mistral 7B around 0.37. Here, the smallest Llama model is both accurate enough and substantially fairest.

For **Bank Marketing**, all Llama and Mistral models exhibit DP close to 1.0, indicating maximal disparity and very poor group fairness on this dataset. Scaling within a family does not help: every model is equally unfair.

On **COMPAS**, the differences are more nuanced. Mistral 7B achieves the lowest DP (approximately 0.22), slightly better than Llama 8B (around 0.23). Llama 405B and Mistral 24B both sit around 0.25–0.28, while Llama 70B is on the higher end (approximately 0.28). No single family dominates; smaller and mid-sized models can be fairest.

For **Diabetes Readmission**, all models—across both families and all sizes—show very low DP (roughly 0.01–0.04), indicating high fairness. Here, the task itself appears easier to satisfy parity conditions, and scaling has little effect.

On **Folktables**, the Mistral models generally achieve better DP than Llama. Mistral 24B has DP around 0.27 and Mistral 7B around 0.28. In contrast, Llama 8B is around 0.35, Llama 405B around 0.40, and Llama 70B is notably higher at approximately 0.65. In this setting, both Mistral models outperform all Llama sizes from a DP perspective, and scaling within Llama substantially worsens fairness.

Finally, for **German Credit**, Mistral 7B achieves the best DP (approximately 0.12). Llama 405B, Llama 70B, and Mistral 24B cluster slightly higher around 0.15, while Llama 8B is less fair at about 0.22. Here, the smaller Mistral model is the fairest, while scaling within Llama does not yield clear gains.

Summary. Overall, the results indicate that:

- Model size *correlates* with higher accuracy on average, but this relationship is not strictly monotonic: Llama 405B is less accurate than Llama 70B.
- Model size does not reliably improve fairness. Within Llama, the smallest model (8B) is typically the fairest, and fairness degrades when moving to 70B and only partially recovers at 405B. Within Mistral, fairness differences between 7B and 24B are small and depend on whether DP or EO is considered.
- Dataset characteristics strongly mediate fairness behavior. Some tasks (e.g., Diabetes Readmission) are fair under all models, while others (e.g., Bank Marketing, Folktables) exhibit large disparities regardless of scale or even worsening with larger models.

In summary, scaling within a model family tends to increase predictive performance but does not provide a consistent path to improved group fairness. In some cases, larger models are less fair than their smaller counterparts, underscoring the need for explicit fairness-aware training or post-hoc mitigation rather than relying on scale alone.

5.2 RQ2: How does fairness vary across different capability tiers?

Our second research question examines whether higher-capability commercial models behave more fairly than their lower-tier counterparts within the same family. We evaluate two capability pairs: Gemini 2.5 Flash vs. Gemini 2.5 Pro

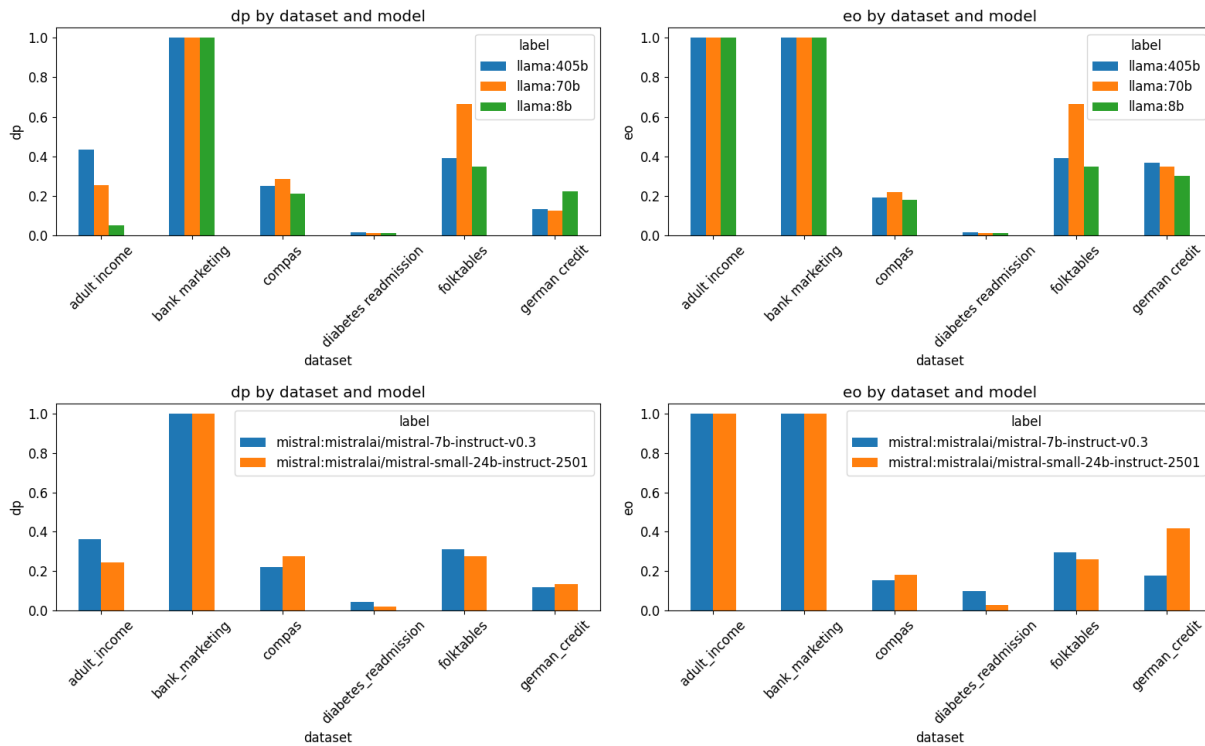


Figure 4: Fairness scaling across Llama and Mistral model families, measured using Demographic Parity (DP) and Equalized Odds (EO) across datasets. Lower values indicate more equitable performance.

and GPT-4o-mini vs. GPT-4o. These pairs allow us to examine fairness differences attributable to capability tier while holding the overall architecture family constant.

Figures 5 and 6 summarize average accuracy, DP, and EO across all datasets, while Figure 7 presents dataset-level DP and EO. Together, these reveal a clear pattern: accuracy improves with capability tier, but fairness outcomes vary substantially by dataset and model family.

Accuracy Effects of Capability Tier. Across both families, higher-tier models achieve higher accuracy. Gemini 2.5 Pro outperforms Flash (0.61 vs. 0.52), and GPT-4o surpasses GPT-4o-mini (0.65 vs. 0.56). Accuracy therefore correlates positively with capability tier, although this carries no guarantee of improved fairness.

Dataset-Level Fairness Patterns. Fairness varies substantially across datasets, and the direction of these changes does not follow a simple capability-based trend.

On **Adult Income**, GPT-4o achieves the lowest DP (0.25), followed closely by Gemini 2.5 Pro (0.26). Gemini Flash exhibits slightly higher disparity (0.28), while GPT-4o-mini shows the highest DP (0.32). Here, the higher-tier models demonstrate modest fairness advantages, although the differences are relatively small.

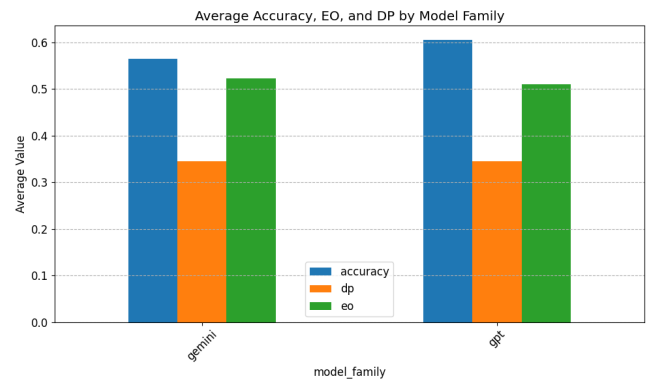


Figure 5: Gemini models: average accuracy, Demographic Parity (DP), and Equalized Odds (EO) aggregated across Gemini 2.5 Flash and 2.5 Pro. Lower DP and EO indicate more equitable behavior across sensitive groups.

For **Bank Marketing**, all models across both families produce DP values of 1.0, indicating maximal disparity in positive prediction rates. This dataset exhibits universally poor fairness performance, and capability tier has no influence: every model is equally unfair.



Figure 6: GPT models: average accuracy, Demographic Parity (DP), and Equalized Odds (EO) aggregated across GPT-4o-mini and GPT-4o. Lower DP and EO indicate more equitable behavior across sensitive groups.

On **COMPAS**, GPT-4o again achieves the lowest DP (0.27), with Gemini Flash close behind at 0.28. GPT-4o-mini follows at 0.29, while Gemini Pro produces the highest disparity at 0.32. These results indicate that neither capability tier nor model family systematically predicts fairness: lower-tier Gemini Flash outperforms higher-tier Gemini Pro, and GPT models show only slight tier-based improvements.

For **Diabetes Readmission**, all models exhibit extremely low DP values. GPT-4o and GPT-4o-mini show the lowest disparities (0.005), while Gemini Pro (0.01) and Gemini Flash (0.02) also perform very well. This task appears inherently easy with respect to demographic parity, and capability tier has minimal impact.

On **Folktables**, Gemini Flash demonstrates substantially better fairness than all competing models, achieving a DP of 0.04. In contrast, Gemini Pro, GPT-4o, and GPT-4o-mini all produce nearly identical and higher disparities of 0.12. This dataset sharply distinguishes between capability tiers: the lower-tier Gemini model is markedly fairer than both its higher-tier version and the GPT series.

For **German Credit**, Gemini Flash again achieves the lowest DP (0.08), followed by GPT-4o-mini (0.10). Gemini Pro shows a higher disparity (0.18), and GPT-4o exhibits the highest DP (0.24). Similar to Folktables, lower-tier models produce the fairest outcomes, while higher-tier versions worsen fairness.

Taken together, these dataset-specific results show that fairness is highly context-dependent and does not follow a simple scaling law. Higher-tier models are occasionally fairer (e.g., Adult Income), but in several cases—particularly Folktables and German Credit—the lower-tier models offer substantially more equitable behavior.

Dataset-Specific Fairness Comparisons (EO). Equalized Odds patterns (Figure 7, right panel) reinforce the conclusion that fairness does not improve reliably with capability tier. GPT-4o generally exhibits higher EO disparities than GPT-4o-mini despite superior accuracy, while Gemini Flash and Pro alternate depending on the dataset. As with DP, the Diabetes task yields uniformly low disparities across all models, whereas Bank Marketing produces uniformly high EO.

Interpretation. Taken together, the DP and EO results show that fairness outcomes are highly dataset-dependent and do not follow capability tier. Higher-tier models improve accuracy but do not consistently reduce disparities; in several datasets—most notably Folktables and German Credit—the lower-tier Gemini Flash is the fairest model. Similarly, GPT-4o-mini often achieves lower disparities than GPT-4o despite being less accurate.

Summary. Overall, capability tier reliably enhances predictive performance but does not guarantee more equitable behavior. Lower-tier models can be as fair as or fairer than their higher-tier counterparts, highlighting the need to evaluate fairness directly rather than inferring it from model capability.

5.3 Summary Table of Model-Level Performance

For convenience, Table 4 summarizes the average performance of each model across all six datasets.

Family	Model	Accuracy	DP	EO
Gemini	2.5 Flash	0.52	0.28	0.44
Gemini	2.5 Pro	0.61	0.30	0.45
GPT	4o-mini	0.56	0.31	0.46
GPT	4o	0.65	0.31	0.51
Llama	8B	0.37	0.31	0.47
Llama	70B	0.53	0.39	0.54
Llama	405B	0.45	0.37	0.49
Mistral	7B	0.43	0.34	0.45
Mistral	24B	0.49	0.32	0.48

Table 4: Average performance across all datasets for each model, based on the summary computed in the analysis notebook. Higher accuracy and lower DP and EO are better.

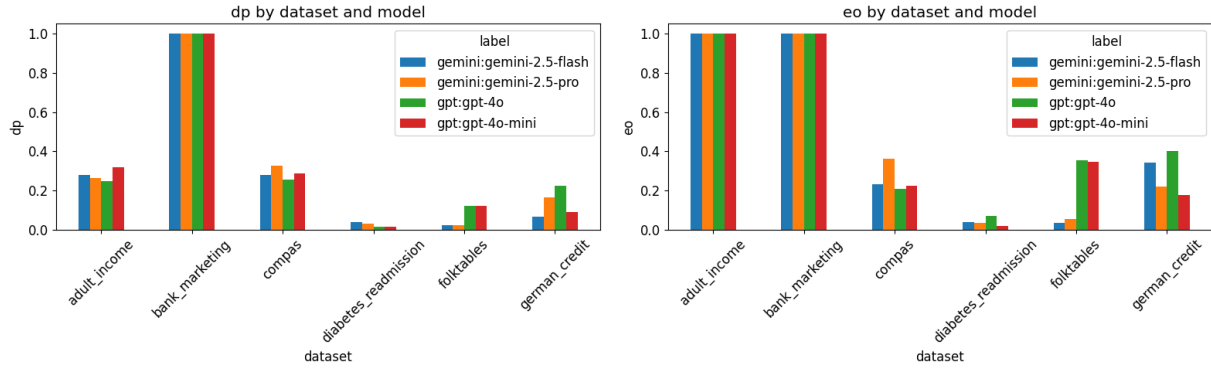


Figure 7: Fairness across commercial capability tiers. Left: Demographic Parity (DP) by dataset for Gemini 2.5 Flash/Pro and GPT-4o/4o-mini. Right: Equalized Odds (EO) for the same models. Lower values indicate greater fairness.

5.4 Conclusion of Empirical Results

Across both research questions, our empirical findings show that:

- Increasing model size within a family reliably improves accuracy but has mixed effects on fairness, sometimes amplifying disparities (RQ1).
- Higher commercial capability tiers offer substantial accuracy gains but do not guarantee better fairness, and in some cases slightly worsen error-rate disparities (RQ2).

These results suggest that practitioners cannot rely on the model size or the capability tier alone as a proxy for fairness. Systematic audit with group-based metrics remains necessary even for large-scale high-performing LLMs.

6 CONCLUSION

This capstone investigated how design choices in modern LLM systems, specifically parameter scale and commercial capability tier, shape both accuracy and algorithmic fairness on high-stakes tabular prediction tasks. Using six real-world datasets from finance, criminal justice, healthcare, and economic opportunity, we prompted Llama, Mistral, Gemini, and GPT models as classifiers, collected their predictions and reasoning traces, and evaluated them with group-based metrics via Fairlearn’s *MetricFrame*. Across families, larger and higher-tier models consistently delivered higher accuracy, but their fairness behavior was mixed: Llama became less fair as it scaled, Mistral showed metric-dependent effects, and higher-tier commercial models such as GPT-4o achieved the strongest accuracy while sometimes exhibiting larger Equalized Odds disparities than their lower-tier counterparts. Overall, neither parameter count nor commercial capability tier emerged as a reliable proxy for equitable behavior.

Methodologically, this work demonstrates a practical framework for auditing LLM fairness on structured data by converting tabular records into standardized prompts, retaining sensitive attributes for evaluation, and computing disaggregated fairness metrics across demographic groups. Reasoning traces offer an additional interpretability layer and create a foundation for future work that analyzes how models justify decisions and when they rely on proxy features. Important limitations remain: we evaluate a single prompting strategy, focus on a subset of fairness metrics compatible with label-only outputs, and consider only a slice of the rapidly evolving LLM ecosystem. These constraints point to clear next steps, exploring fairness-aware prompting, richer metric families, causal or individual-level analyses, and broader model coverage. The central message, however, is clear: as LLMs become more capable, their fairness properties do not automatically improve and may in fact worsen, making systematic, dataset-aware fairness auditing an essential requirement for any deployment in high-stakes domains.

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